A COMPUTER-FREE EXCLUSION OF AUTOMORPHISMS OF ORDER 7 IN A PUTATIVE srg(99, 14, 1, 2)

CONWAY99-STRUCTURES-LAB

Abstract. We prove that a strongly regular graph with parameters $(99, 14, 1, 2)$ (a *Conway 99-graph*) has no automorphism of order 7. Together with the computer-free reduction of Cesarz and Woldar, this shows without computer assistance that the order of the automorphism group of such a graph is not divisible by 7; previously this was known through the orbit-matrix computations of Behbahani and Lam. An automorphism of order 7 yields a symmetric nonnegative integral 12×12 matrix C (the exterior orbit quotient) satisfying $C^2 + C = 12I + 12J - 2uu^{\mathsf{T}}$, $C\mathbf{1} = 12\mathbf{1}$, $Cu = 0$. A rational-representation argument forces C to have zero diagonal. Then $G = 3C + 12I - 4J - 2uu^{\mathsf{T}}$ is 21 times a projection of rank 4, so C is encoded by twelve vectors in $\mathbb{R}^4$. A short case analysis, carried out entirely by hand, shows that such vectors cannot exist.

Status. The strategy (exterior quotient matrix, trace argument, rank-4 Gram compression, case analysis on the LL/RR block) comes from a candidate proof that an AI system (ChatGPT) produced on 2 October 2026. This text re-derives every step. It replaces the computer-assisted finite checks of that candidate with hand derivations (Sections 7 and 8) and adds independent exact verifications (Section 9). A Lean 4 formalization is reported in Section 9. The argument has not yet been refereed.

1. Introduction

Conway asked whether a strongly regular graph with parameters $(99, 14, 1, 2)$ exists: a graph on 99 vertices in which every edge lies in exactly one triangle and every non-edge is a diagonal of exactly one quadrilateral [4]. The question is open. A standard line of attack is to constrain the automorphism group of a putative example. Behbahani and Lam [1] used orbit matrices, with computer assistance, to rule out automorphisms of several prime orders, including 7. Cesarz and Woldar [3] gave computer-free proofs that if 7 divides $|\operatorname{Aut}\Gamma|$ then $\operatorname{Aut}\Gamma \cong \mathbb{Z}_7$, and that if 2 divides $|\operatorname{Aut}\Gamma|$ then $|\operatorname{Aut}\Gamma|$ divides 6. The present note completes the computer-free picture for the prime 7.

Theorem 1.1. *If Γ is a strongly regular graph with parameters $(99, 14, 1, 2)$, then Γ has no automorphism of order 7. Consequently $7 \nmid |\operatorname{Aut}\Gamma|$.*

The proof only uses the existence of a single automorphism s of order 7. It does not use the structure of the full automorphism group. We reprove the orbit lemma (Lemma 3.1) instead of quoting it.

Plan. Section 2 recalls the local structure of Γ. Section 3 describes the orbits of s, and Section 4 derives the matrix equations for the exterior quotient C. Section 5 shows C has zero diagonal (the only representation-theoretic input). Section 6 introduces the rank-4 Gram vectors. Section 7 classifies the 6×6 block of C on the LL and RR orbits (six cases), and Section 8 kills each case using the mixed orbits.

2. Preliminaries

Let $\Gamma = (V, E)$ be an srg$(99, 14, 1, 2)$ with adjacency matrix A, and write J for the all-ones matrix. Then $A^2 = 14I + A + 2(J - I - A)$, that is,

$$A^2 + A = 12I + 2J. \tag{1}$$

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The eigenvalues are 14 (on $\mathbf{1}$) and the roots $3, -4$ of $X^2 + X - 12$. Their multiplicities f, g satisfy $f + g = 98$ and $14 + 3f - 4g = \operatorname{tr} A = 0$, so the spectrum is $14^1\, 3^{54}\, (-4)^{44}$.

For a vertex x write $\Gamma(x)$ for its neighbourhood and $\Gamma_2(x) = V \setminus (\{x\} \cup \Gamma(x))$ ($|\Gamma_2(x)| = 84$).

Lemma 2.1 (local structure)**.** *Let $x \in V$.*

(a) $\Gamma(x)$ induces a perfect matching $7K_2$. For $a \in \Gamma(x)$ write a' for its partner.
(b) Every $y \in \Gamma_2(x)$ has exactly two neighbours in $\Gamma(x)$, and they are non-adjacent.
(c) The map $y \mapsto \Gamma(y) \cap \Gamma(x)$ is a bijection from $\Gamma_2(x)$ onto the 84 non-adjacent pairs of $\Gamma(x)$.
(d) Every vertex of $\Gamma(x)$ has 12 neighbours in $\Gamma_2(x)$.

Proof. (a) Since $\lambda = 1$, every $a \in \Gamma(x)$ has exactly one neighbour in $\Gamma(x)$. (b) Since $\mu = 2$, y has exactly two neighbours $a, b \in \Gamma(x)$. If $a \sim b$, then a, b would have the two common neighbours x, y, contradicting $\lambda = 1$. (c) Let $a, b \in \Gamma(x)$ be non-adjacent. They have $\mu = 2$ common neighbours, one of which is x. The other is not in $\Gamma(x)$, because the only neighbour of a in $\Gamma(x)$ is a', and $a' \neq b'$. So exactly one $y \in \Gamma_2(x)$ is adjacent to both. Together with (b) this gives the bijection, and $\binom{14}{2} - 7 = 84 = |\Gamma_2(x)|$. (d) $14 - 1 - 1 = 12$. □

3. The orbits of an automorphism of order 7

From now on $s \in \operatorname{Aut}\Gamma$ has order 7. Every orbit of $\langle s \rangle$ has size 1 or 7.

Lemma 3.1 (cf. [3])**.** *s fixes exactly one vertex x, and s fixes no vertex of $\Gamma(x)$ or of $\Gamma_2(x)$.*

Proof. Since $99 \equiv 1 \pmod 7$, the number of fixed vertices is $\equiv 1 \pmod 7$; in particular some vertex x is fixed. The set F of fixed vertices in $\Gamma(x)$ is closed under $a \mapsto a'$ (the partner is defined in terms of x and a), and $|F| \equiv |\Gamma(x)| = 14 \equiv 0 \pmod 7$. So $|F| \in \{0, 14\}$, as $|F|$ is even. If $|F| = 14$, every $y \in \Gamma_2(x)$ is fixed, because by Lemma 2.1(c) it is determined by its two (fixed) neighbours in $\Gamma(x)$. Then $s = 1$, a contradiction. Hence $F = \emptyset$.

If $y \in \Gamma_2(x)$ were fixed, s would fix its two-element set $\Gamma(y) \cap \Gamma(x)$ setwise. Since s has odd order, it would fix both points, which contradicts $F = \emptyset$. Every vertex is at distance at most 2 from x, so x is the only fixed vertex. □

Lemma 3.2. *The vertices of $\Gamma(x)$ can be labelled ℓ_i, r_i ($i \in \mathbb{Z}/7$) so that $\ell_i \sim r_i$, $s(\ell_i) = \ell_{i+1}$ and $s(r_i) = r_{i+1}$.*

Proof. s permutes the seven matching edges of $\Gamma(x)$. If s fixed an edge $\{a, a'\}$ setwise, then s^2 would fix a, hence so would $s = (s^2)^4$; this is impossible. So the matching edges form a single $\langle s \rangle$-orbit. Choose $\ell_0 \in \Gamma(x)$, put $\ell_i = s^i(\ell_0)$ and $r_i = \ell_i'$. Then $s(r_i) = s(\ell_i)' = r_{i+1}$. The edges $\{\ell_i, r_i\} = s^i\{\ell_0, r_0\}$ are the seven distinct matching edges, so the 14 labels are distinct. □

Write $L = \{\ell_i\}$ and $R = \{r_i\}$. By Lemma 2.1(c), each $y \in \Gamma_2(x)$ is labelled by a non-adjacent pair of $\Gamma(x)$, and s acts on labels through $i \mapsto i + 1$. The exterior $\Gamma_2(x)$ therefore splits into twelve orbits of size 7:

- three *LL-orbits* (labels $\{\ell_i, \ell_{i+\delta}\}$, $\delta = 1, 2, 3$), forming the set $\mathcal{A}$;
- three *RR-orbits* (labels $\{r_i, r_{i+\delta}\}$, $\delta = 1, 2, 3$), forming the set $\mathcal{B}$;
- six *mixed orbits* (labels $\{\ell_i, r_{i+\delta}\}$, $\delta = 1, \dots, 6$), forming the set $\mathcal{M}$.

For an exterior orbit O put $u_O = 1, -1, 0$ according as $O \in \mathcal{A}, \mathcal{B}, \mathcal{M}$. A vertex of O then has $1 + u_O$ neighbours in L and $1 - u_O$ neighbours in R.

4. The exterior quotient matrix

For exterior orbits O, O' let $C_{OO'}$ be the number of neighbours in O' of a vertex $y \in O$. This does not depend on $y \in O$, because s is an automorphism. Order the orbits as $\mathcal{A}, \mathcal{B}, \mathcal{M}$, so that

$$u = (1, 1, 1, -1, -1, -1, 0, 0, 0, 0, 0, 0)^{\mathsf{T}}.$$

Proposition 4.1. *C is a symmetric 12×12 matrix with nonnegative integer entries, and*

$$C\mathbf{1} = 12\,\mathbf{1}, \tag{2}$$

$$Cu = 0, \tag{3}$$

$$C^2 + C = 12I + 12J - 2uu^{\mathsf{T}}. \tag{4}$$

Proof. Symmetry holds because both $7C_{OO'}$ and $7C_{O'O}$ count the edges between O and O'. For (2): a vertex $y \in \Gamma_2(x)$ has 14 neighbours, two of them in $\Gamma(x)$ (Lemma 2.1) and none equal to x. A vertex of L has exactly $1 + u_{O'}$ neighbours in O', since both counts equal $1/7$ of the number of edges between L and O'; likewise $1 - u_{O'}$ for R.

Fix $y \in O$ and count the walks $y \to w \to z$ with z in a fixed set Z. Grouping by z gives $\sum_{z\in Z} |\Gamma(y) \cap \Gamma(z)|$, and $|\Gamma(y) \cap \Gamma(z)|$ is $14, 1, 2$ according as $z = y$, $z \sim y$, or neither. Grouping by w gives $\sum_{w\sim y} |\Gamma(w) \cap Z|$.

For $Z = O'$ the first count is $14[O{=}O'] + C_{OO'} + 2(7 - [O{=}O'] - C_{OO'}) = 12[O{=}O'] + 14 - C_{OO'}$. The second is

$$(1 + u_O)(1 + u_{O'}) + (1 - u_O)(1 - u_{O'}) + \sum_{O''} C_{OO''}C_{O''O'} = 2 + 2u_O u_{O'} + (C^2)_{OO'}.$$

Equating the two gives (4).

For $Z = L$ the first count is $(1 + u_O) + 2(6 - u_O) = 13 - u_O$. In the second, neighbours $w \in L$ contribute 0, since L is independent. Neighbours $w \in R$ contribute $(1 - u_O) \cdot 1$. Exterior neighbours contribute $\sum_{O''} C_{OO''}(1 + u_{O''}) = 12 + (Cu)_O$. Hence $(Cu)_O = 0$. □

Taking diagonal entries in (4):

$$\sum_j C_{ij}^2 + C_{ii} = 24 - 2u_i^2. \tag{5}$$

Lemma 4.2 (spectrum of C)**.** *$C\mathbf{1} = 12\mathbf{1}$ and $Cu = 0$. On $W := \langle \mathbf{1}, u\rangle^{\perp}$, which is C-invariant and of dimension 10, the only eigenvalues of C are 3 and -4. If a denotes the multiplicity of 3 on W, then $\operatorname{tr} C = 7a - 28$.*

Proof. W is invariant because C is symmetric and $\mathbf{1}, u$ are eigenvectors. For $w \in W$, (4) gives $(C^2 + C)w = 12w$, so the eigenvalues there are roots of $X^2 + X - 12$. Finally $\operatorname{tr} C = 12 + 0 + 3a - 4(10 - a)$. □

5. The spectral input: C has zero diagonal

Lemma 5.1. *Let E be a finite-dimensional $\mathbb{Q}$-vector space and $T \in \mathrm{GL}(E)$ with $T^7 = 1$. Then $\dim E - \dim\ker(T - 1) \equiv 0 \pmod 6$.*

Proof. $X^7 - 1 = (X - 1)\Phi_7(X)$ with coprime factors, so $E = \ker(T - 1) \oplus E'$ with $E' = \ker \Phi_7(T)$. The polynomial Φ_7 is irreducible over $\mathbb{Q}$ (Eisenstein at 7 after $X \mapsto X + 1$). Hence $K = \mathbb{Q}[X]/(\Phi_7)$ is a field of degree 6, and E' is a K-vector space with X acting as T. Therefore $\dim_{\mathbb{Q}} E' = 6 \dim_K E'$. □

Proposition 5.2. *$\operatorname{tr} C = 0$. Hence $C_{ii} = 0$ for all i, and*

$$\sum_j C_{ij} = 12, \qquad \sum_j C_{ij}^2 = \begin{cases} 22, & i \in \mathcal{A} \cup \mathcal{B}, \\ 24, & i \in \mathcal{M}. \end{cases} \tag{6}$$

In particular every entry of C is at most 4.

Proof. Let P be the permutation matrix of s, and let $F = \{f \in \mathbb{R}^V : Pf = f\}$, the span of the 15 orbit indicators. Since A commutes with P, A preserves F. In the basis of orbit indicators, $A|_F$ has matrix B, where $B_{OO'}$ is the number of neighbours in O' of a vertex of O. Its diagonal entries are 0 for $\{x\}$, L and R (L and R are independent sets), so $\operatorname{tr}(A|_F) = \operatorname{tr} C$.

$A|_F$ is self-adjoint, hence diagonalizable, with eigenvalues among $14, 3, -4$. The eigenvalue 14 has multiplicity 1, because $\mathbf{1} \in F$. Let $d = \dim(F \cap E_3)$, where $E_3 = \ker(A - 3I)$. Then -4 has multiplicity $14 - d$, and

$$\operatorname{tr} C = \operatorname{tr}(A|_F) = 14 + 3d - 4(14 - d) = 7d - 42.$$

A and P are rational matrices, so E_3 has a rational form $E_{3,\mathbb{Q}} = \ker_{\mathbb{Q}}(A-3I)$ of $\mathbb{Q}$-dimension 54. It is P-invariant, and $\ker(P-1) \cap E_{3,\mathbb{Q}}$ has $\mathbb{Q}$-dimension d. By Lemma 5.1, $54 - d \equiv 0 \pmod 6$. Since $0 \le d \le 14$, $d \in \{0, 6, 12\}$.

If $d = 0$ then $\operatorname{tr} C = -42 < 0$, which is impossible. If $d = 12$ then $\operatorname{tr} C = 42$, and Lemma 4.2 gives $a = 10$. So C acts as 3 on all of W:

$$C = 12 \cdot \tfrac{1}{12} J + 0 \cdot \tfrac{1}{6} uu^{\mathsf{T}} + 3(I - \tfrac{1}{12} J - \tfrac{1}{6} uu^{\mathsf{T}}) = 3I + \tfrac{3}{4} J - \tfrac{1}{2} uu^{\mathsf{T}}.$$

Then $C_{mm'} = 3/4$ for distinct $m, m' \in \mathcal{M}$, which is not an integer. Hence $d = 6$ and $\operatorname{tr} C = 0$. Since $C \ge 0$ entrywise, the diagonal vanishes, and (6) follows from (2) and (5). □

From now on $a = d - 2 = 4$.

6. Gram compression

Put

$$G = 3C + 12I - 4J - 2uu^{\mathsf{T}}. \tag{7}$$

G vanishes on $\mathbf{1}$ and u; on W it is $3 \cdot 3 + 12 = 21$ on the 3-eigenspace of C and $3 \cdot (-4) + 12 = 0$ on the (-4)-eigenspace. So G is 21 times the orthogonal projection onto a 4-dimensional space:

$$G \succeq 0, \qquad G^2 = 21G, \qquad \operatorname{rank} G = 4. \tag{8}$$

Hence there are vectors $v_1, \dots, v_{12} \in \mathbb{R}^4$ spanning $\mathbb{R}^4$ with $G_{ij} = v_i \cdot v_j$. Explicitly,

$$|v_i|^2 = \begin{cases} 6, & i \in \mathcal{A} \cup \mathcal{B} \\ 8, & i \in \mathcal{M} \end{cases} \qquad v_i \cdot v_j = \begin{cases} 3C_{ij} - 6, & i,j \in \mathcal{A} \text{ or } i,j \in \mathcal{B}, \\ 3C_{ij} - 2, & i \in \mathcal{A},\ j \in \mathcal{B}, \\ 3C_{ij} - 4, & i \text{ or } j \in \mathcal{M} \end{cases} \quad (i \ne j). \tag{9}$$

Moreover $G\mathbf{1} = 0$ and $Gu = 0$ translate into

$$\sum_i v_i = 0, \qquad \sum_{a \in \mathcal{A}} v_a = \sum_{b \in \mathcal{B}} v_b. \tag{10}$$

Write $\mathcal{A} = \{a_1, a_2, a_3\}$ and $\mathcal{B} = \{b_1, b_2, b_3\}$. Let $P = (C_{a_i a_j})$, $Q = (C_{b_i b_j})$ and $R = (C_{a_i b_j})$ be the 3×3 blocks. For $a \in \mathcal{A}$, row a of (3) says $\sum_{a' \in \mathcal{A}} C_{aa'} = \sum_{b \in \mathcal{B}} C_{ab} =: \sigma_a$; that is,

$$P\mathbf{1} = R\mathbf{1}, \qquad Q\mathbf{1} = R^{\mathsf{T}}\mathbf{1}. \tag{11}$$

In particular the three *edge weights* of P and of Q have the same sum, which we call h.

7. The LL/RR block

Lemma 7.1. *$C_{aa'} \ge 1$ for distinct $a, a' \in \mathcal{A}$, and likewise in $\mathcal{B}$.*

Proof. If $C_{aa'} = 0$ then $|v_a + v_{a'}|^2 = 6 + 6 - 12 = 0$, so $v_{a'} = -v_a$. For $b \in \mathcal{B}$ this gives $(3C_{ab} - 2) + (3C_{a'b} - 2) = 0$, i.e. $3(C_{ab} + C_{a'b}) = 4$, which is impossible. The same argument works in $\mathcal{B}$. □

Lemma 7.2. *$h \in \{3, 4, 5\}$.*

Proof. Let $e_{\mathcal{A}}, e_{\mathcal{B}}$ be the indicator vectors and $t = Ce_{\mathcal{A}}$; by (3), $t = Ce_{\mathcal{B}}$ as well. Then $\sum_{\mathcal{A}} t_i = e_{\mathcal{A}}^{\mathsf{T}} C e_{\mathcal{A}} = 2h$, $\sum_{\mathcal{B}} t_i = e_{\mathcal{B}}^{\mathsf{T}} C e_{\mathcal{A}} = e_{\mathcal{B}}^{\mathsf{T}} C e_{\mathcal{B}} = 2h$, and $\sum_{\mathcal{M}} t_i = \mathbf{1}^{\mathsf{T}} C e_{\mathcal{A}} - 4h = 36 - 4h$. By (4),

$$|t|^2 = e_{\mathcal{A}}^{\mathsf{T}} C^2 e_{\mathcal{A}} = e_{\mathcal{A}}^{\mathsf{T}}(12I + 12J - 2uu^{\mathsf{T}} - C)e_{\mathcal{A}} = 36 + 108 - 18 - 2h = 126 - 2h.$$

Applying Cauchy–Schwarz on the blocks $\mathcal{A}, \mathcal{B}, \mathcal{M}$ (of sizes $3, 3, 6$) gives $126 - 2h \ge \frac{(2h)^2}{3} + \frac{(2h)^2}{3} + \frac{(36-4h)^2}{6}$, which simplifies to $(h-3)(8h-45) \le 0$. □

By Lemmas 7.1 and 7.2, the edge weights of P (and of Q) form one of the multisets $\{1,1,1\}$ ($h=3$), $\{1,1,2\}$ ($h=4$), $\{1,1,3\}$ or $\{1,2,2\}$ ($h=5$). Relabelling $\mathcal{A}$ and $\mathcal{B}$ separately, we may take P and Q to be one of

$$T_{111}=\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix},\ T_{112}=\begin{pmatrix}0&1&1\\1&0&2\\1&2&0\end{pmatrix},\ T_{113}=\begin{pmatrix}0&1&1\\1&0&3\\1&3&0\end{pmatrix},\ T_{122}=\begin{pmatrix}0&1&2\\1&0&2\\2&2&0\end{pmatrix}.$$

7.1. **The case $h=3$.**

Proposition 7.3. *If $h=3$ then $P=Q=T_{111}$, $\sum_{\mathcal{A}} v_a=\sum_{\mathcal{B}} v_b=0$, and all row and column sums of R equal 2. Up to relabelling, R is one of*

$$R_1=2I,\qquad R_2=\begin{pmatrix}2&0&0\\0&1&1\\0&1&1\end{pmatrix},\qquad R_3=T_{111}.$$

Moreover every mixed m satisfies $\sum_{a\in\mathcal{A}} C_{ma}=\sum_{b\in\mathcal{B}} C_{mb}=4$.

Proof. $|\sum_{\mathcal{A}} v_a|^2=3\cdot 6+6\cdot(-3)=0$, and similarly for $\mathcal{B}$. Pairing $\sum_{\mathcal{A}} v_a=0$ with v_b gives $\sum_a(3R_{ab}-2)=0$, so the column sums of R are 2; the row sums follow in the same way. Pairing with v_m gives $\sum_a(3C_{ma}-4)=0$. A nonnegative integral 3×3 matrix with all line sums 2 is a sum of two permutation matrices $\pi_1+\pi_2$, and up to row and column permutations $\pi_1^{-1}\pi_2$ is the identity, a transposition, or a 3-cycle; these give R_1,R_2,R_3. All such relabellings preserve $P=Q=T_{111}$. □

7.2. **The cases $h=4,5$: residuals.** Now $P\in\{T_{112},T_{113},T_{122}\}$, and its Gram block $G_{\mathcal{A}}=(v_a\cdot v_{a'})$ has $\det G_{\mathcal{A}}=54(xyz-(h-4)^2)>0$, where x,y,z are the edge weights. So v_{a_1},v_{a_2},v_{a_3} span a 3-space $U\subset\mathbb{R}^4$. Fix a unit normal n of U and put $z_b=v_b\cdot n$ for $b\in\mathcal{B}$. Write $c=(R_{a_1b},R_{a_2b},R_{a_3b})$ for the column of R at b, and $X=3c-\mathbf{21}$ for the inner products of v_b with the v_a. Then

$$z_b^2=6-X^{\mathsf{T}}G_{\mathcal{A}}^{-1}X,\qquad z_{b_1}+z_{b_2}+z_{b_3}=0, \tag{12}$$

the second identity because $\sum_{\mathcal{B}} v_b=\sum_{\mathcal{A}} v_a\in U$ by (10). The column sum of c is σ_b, the row sum of Q at b, by (11). Three real numbers with zero sum have the largest absolute value equal to the sum of the other two. So if $\alpha_b=12z_b^2$, then $\sqrt{\alpha_{\max}}=\sqrt{\alpha_j}+\sqrt{\alpha_k}$ for the two others, and in particular $\alpha_j\alpha_k$ is a perfect square. The following tables list $\alpha=12z^2$ for every column c with the required sum and $z^2\ge 0$. They follow from (12) with

$$12\,G_{T_{112}}^{-1}=\begin{pmatrix}4&2&2\\2&3&1\\2&1&3\end{pmatrix},\quad 12\,G_{T_{113}}^{-1}=\begin{pmatrix}3&1&1\\1&3&-1\\1&-1&3\end{pmatrix},\quad 12\,G_{T_{122}}^{-1}=\tfrac{2}{3}\begin{pmatrix}4&2&0\\2&4&0\\0&0&3\end{pmatrix}.$$

P	column sum	columns c with $\alpha=12z^2$
T_{112}	2	$(0,1,1)$: 64; $(1,0,1),(1,1,0)$: 61; $(2,0,0)$: 40; $(0,0,2),(0,2,0)$: 28
	3	$(1,1,1)$: 52; $(0,1,2),(0,2,1)$: 37; $(1,0,2),(1,2,0)$: 16; $(2,0,1),(2,1,0)$: 13
T_{113}	2	$(0,1,1)$: 64; $(1,0,1),(1,1,0)$: 52; $(2,0,0)$: 40
	3	$(1,1,1)$: 61; $(0,1,2),(0,2,1)$: 37; $(2,0,1),(2,1,0)$: 13
	4	$(0,2,2)$: 28; $(1,1,2),(1,2,1)$: 16; $(2,1,1)$: 4
T_{122}	3	$(1,1,1)$: 62; $(0,2,1),(2,0,1)$: 38; $(0,1,2),(1,0,2)$: 32; $(1,2,0),(2,1,0)$: 8
	4	$(1,1,2)$: 32; $(1,2,1),(2,1,1)$: 14; $(0,2,2),(2,0,2)$: 8

Columns not listed have $z^2<0$. For T_{112} and T_{113} the row sums are $(\sigma_{a_1},\sigma_{a_2},\sigma_{a_3})=(2,3,3)$ and $(2,4,4)$; for T_{122} they are $(3,3,4)$, with a_3 the vertex of the two 2-edges.

Proposition 7.4. *If $h \in \{4,5\}$ then $P = Q$ up to relabelling, and up to relabelling (P, R) is one of*

$$(T_{112}, T_{112}), \qquad \Big(T_{113}, \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}\Big), \qquad \Big(T_{122}, \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}\Big).$$

Proof. We use the tables with the α's of the three columns.

$P = Q = T_{112}$. The column at b_1 has sum $\sigma_{b_1} = 2$, and those at b_2, b_3 have sum 3, so $\alpha_{b_1} \in \{64, 61, 40, 28\}$ and $\alpha_{b_2}, \alpha_{b_3} \in \{52, 37, 16, 13\}$. For each pair of columns whose values have a square product, the remaining column would need the value $(\sqrt{\alpha_j} + \sqrt{\alpha_k})^2$. For the pair (b_2, b_3) the square products are $16 \cdot 16$, $13 \cdot 13$, $13 \cdot 52$, $52 \cdot 52$ and $37 \cdot 37$, which require $\alpha_{b_1} = 64, 52, 117, 208, 148$. For a pair (b_1, b_j) the only square product is $64 \cdot 16$, which requires 144. Only $\sqrt{16} + \sqrt{16} = \sqrt{64}$ is realized, so $\alpha = (64, 16, 16)$: $c_{b_1} = (0, 1, 1)$ and $c_{b_2}, c_{b_3} \in \{(1, 0, 2), (1, 2, 0)\}$. The row sums $(2, 3, 3)$ of R force these two columns to be different. Swapping b_2, b_3 preserves Q, so $R = T_{112}$.

$P = Q = T_{113}$. Here $\alpha_{b_1} \in \{64, 52, 40\}$ (column sum 2) and $\alpha_{b_2}, \alpha_{b_3} \in \{28, 16, 4\}$ (column sum 4). For the pair (b_2, b_3) the square products $16 \cdot 16$, $4 \cdot 4$, $16 \cdot 4$ and $28 \cdot 28$ require $\alpha_{b_1} = 64, 16, 36, 112$. For a pair (b_1, b_j) the square products $64 \cdot 16$ and $64 \cdot 4$ require 144 and 100. Only $\sqrt{16} + \sqrt{16} = \sqrt{64}$ is realized. So $c_{b_1} = (0, 1, 1)$ and $c_{b_2}, c_{b_3} \in \{(1, 1, 2), (1, 2, 1)\}$. The row sums $(2, 4, 4)$ make them different, which gives the stated R up to swapping b_2, b_3.

$P = T_{113}$, $Q = T_{122}$. Q has row sums $(3, 3, 4)$, so $\alpha_{b_1}, \alpha_{b_2} \in \{61, 37, 13\}$ and $\alpha_{b_3} \in \{28, 16, 4\}$. Since $61, 37, 13$ are squarefree and pairwise coprime, the only square products are $61^2, 37^2, 13^2$ for the pair (b_1, b_2). These would require $\alpha_{b_3} = 244, 148, 52$. No product $\alpha_{b_j}\alpha_{b_3}$ with $j \le 2$ is a square. No R exists. Exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$ (that is, replacing u by $-u$, which preserves (2)–(4)) disposes of $P = T_{122}$, $Q = T_{113}$.

$P = Q = T_{122}$. The columns have sums $3, 3, 4$, so $\alpha_{b_1}, \alpha_{b_2} \in \{62, 38, 32, 8\}$ and $\alpha_{b_3} \in \{32, 14, 8\}$. The only square products between different columns are $8 \cdot 8$, $8 \cdot 32$ and $32 \cdot 32$, which require 32, 72 and 128 at the remaining column. So the only valid identity is $\sqrt{8} + \sqrt{8} = \sqrt{32}$. If the 32 sits at b_3, then $c_{b_3} = (1, 1, 2)$ and $c_{b_1}, c_{b_2} \in \{(1, 2, 0), (2, 1, 0)\}$, so row a_3 of R sums to $2 \neq 4$. So the 32 sits at b_1 or b_2; by the symmetry $b_1 \leftrightarrow b_2$ of Q, say at b_1. Then $c_{b_1} \in \{(0, 1, 2), (1, 0, 2)\}$, $c_{b_2} \in \{(1, 2, 0), (2, 1, 0)\}$ and $c_{b_3} \in \{(0, 2, 2), (2, 0, 2)\}$. The row sums $(3, 3, 4)$ leave exactly $R = \left(\begin{smallmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 2 \end{smallmatrix}\right)$. The swap $a_1 \leftrightarrow a_2$ (a symmetry of P) exchanges them. □

Theorem 7.5 (the six configurations)**.** *Up to relabelling $\mathcal{A}$ and $\mathcal{B}$ (and exchanging $\mathcal{A}$ with $\mathcal{B}$), the 6×6 block of C on $\mathcal{A} \cup \mathcal{B}$ is one of the following. In every case $Q = P$ and $R = R^{\mathsf{T}}$.*

case	h	$P = Q$	R
1	*3*	T_{111}	$2I$
2	*3*	T_{111}	$\left(\begin{smallmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$
3	*3*	T_{111}	T_{111}
4	*4*	T_{112}	T_{112}
5	*5*	T_{113}	$\left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{smallmatrix}\right)$
6	*5*	T_{122}	$\left(\begin{smallmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \end{smallmatrix}\right)$

8. The mixed orbits

In each case of Theorem 7.5 put $w_i = v_{a_i} - v_{b_i}$ and $s_i = v_{a_i} + v_{b_i}$ $(i = 1, 2, 3)$.

Lemma 8.1. *$w_i \cdot s_j = 0$ for all i, j. If $\mathsf{W} = \langle w_i \rangle$ and $\mathsf{S} = \langle s_i \rangle$, then $\mathbb{R}^4 = \mathsf{W} \oplus \mathsf{S}$ is an orthogonal decomposition. The Gram matrices are*

$$G^w = 2(3P - 3R + 12I - 4J), \qquad G^s = 2(3P + 3R + 12I - 8J).$$

Proof. $w_i \cdot s_j = (v_{a_i} \cdot v_{a_j} - v_{b_i} \cdot v_{b_j}) + (v_{a_i} \cdot v_{b_j} - v_{b_i} \cdot v_{a_j})$. The first bracket vanishes because $Q = P$, the second because $R = R^{\mathsf{T}}$. The Gram matrices follow from (9). Since $v_{a_i} = \frac{1}{2}(s_i + w_i)$ and $v_{b_i} = \frac{1}{2}(s_i - w_i)$, we get $\mathsf{W} + \mathsf{S} \supseteq \langle v_a, v_b \rangle$. One checks $\operatorname{rank} G^w + \operatorname{rank} G^s = 4$ in each case (see below), so $\mathsf{W} \oplus \mathsf{S} = \mathbb{R}^4$. □

Let $m \in \mathcal{M}$ and write $x_i = C_{ma_i}$, $y_i = C_{mb_i}$ (integers in $[0, 4]$), $d_i = x_i - y_i$, $n_i = x_i + y_i$ and $e_i = 3n_i - 8$. By (9),

$$v_m \cdot w_i = 3d_i, \qquad v_m \cdot s_i = e_i, \qquad |v_m|^2 = 8 = |\pi_{\mathsf{W}} v_m|^2 + |\pi_{\mathsf{S}} v_m|^2. \tag{13}$$

Row m of (3) gives $\sum_i d_i = 0$. Note $d_i \equiv n_i \pmod 2$, $|d_i| \le n_i$, and $e_i \equiv 1 \pmod 3$. For three vectors f_1, f_2, f_3 spanning a plane with Gram matrix $c(3I - J)$ (a "triangle frame"), every z in the plane satisfies $|z|^2 = \frac{1}{3c} \sum_i (z \cdot f_i)^2$.

Case 1. $G^w = 2(3I - J)$ and $G^s = 10(3I - J)$ are triangle frames. In (13) this gives $\frac{3}{2} \sum d_i^2 + \frac{1}{30} \sum e_i^2 = 8$. Since $h = 3$, $\sum s_i = 0$, so $\sum e_i = 0$; put $e_i = 3k_i + 1$ with $\sum k_i = -1$ and $k_i = n_i - 3 \ge -3$. Then $\sum e_i^2 = 9 \sum k_i^2 - 3$ and

$$5 \sum d_i^2 + \sum k_i^2 = 27, \qquad k_i \equiv d_i + 1 \pmod 2.$$

As $\sum d_i = 0$, either all d_i are even or exactly two are odd. If all are even, then $\sum d_i^2 \in \{0, 8, \dots\}$ forces $d = 0$ and $\sum k_i^2 = 27$ with all k_i odd. That means $\{25, 1, 1\}$ or $\{9, 9, 9\}$, and neither allows $\sum k_i = -1$ with $k_i \ge -3$. Otherwise $\sum d_i^2 = 2$, so d is a permutation of $(0, 1, -1)$ and $\sum k_i^2 = 17$, with k_i odd exactly where $d_i = 0$. The only possibility with $\sum k = -1$ is $k = (3, -2, -2)$ in that pattern. Hence $n = (6, 1, 1)$ up to order, and

$$(x; y) \in \{(3, 1, 0; 3, 0, 1),\ (3, 0, 1; 3, 1, 0)\} \quad \text{up to a common permutation of the indices.}$$

In particular $C_{ma_1} \in \{0, 1, 3\}$.

Case 2. W has the orthogonal basis $w_1,\ w_2 - w_3$ (norms $4, 36$), and S has the orthogonal basis $s_1,\ s_2 - s_3$ (norms $20, 36$). So $\frac{9}{4} d_1^2 + \frac{1}{4}(d_2 - d_3)^2 + \frac{1}{20} e_1^2 + \frac{1}{4}(n_2 - n_3)^2 = 8$, i.e.

$$45 d_1^2 + 5(d_2 - d_3)^2 + e_1^2 + 5(n_2 - n_3)^2 = 160.$$

So $5 \mid e_1$, hence $n_1 \in \{1, 6\}$; also $\sum n_i = 8$ (Proposition 7.3). If $n_1 = 6$ then $|d_1| \le 2$ is even and $45 d_1^2 \le 60$ gives $d_1 = 0$. Then $(d_2 - d_3)^2 + (n_2 - n_3)^2 = 12$ with $n_2 + n_3 = 2$, but the left side is at most 8. If $n_1 = 1$ then $d_1 = \pm 1$ and $(d_2 - d_3)^2 + (n_2 - n_3)^2 = 18$ with $n_2 + n_3 = 7$. Both terms are odd squares, so both equal 9. Solving, with the parity constraints, gives

$$(x; y) \in \{(1, 3, 0; 0, 2, 2),\ (1, 0, 3; 0, 2, 2),\ (0, 2, 2; 1, 3, 0),\ (0, 2, 2; 1, 0, 3)\},$$

so $C_{ma_1} \in \{0, 1\}$.

Case 3. $G^w = 8(3I - J)$ and $G^s = 4(3I - J)$ are triangle frames: $\frac{3}{8} \sum d_i^2 + \frac{1}{12} \sum e_i^2 = 8$. With $e_i = 3k_i + 1$ and $\sum k_i = -1$ as in Case 1,

$$\sum d_i^2 + 2 \sum k_i^2 = 22.$$

If exactly two d_i are odd, then $\sum d_i^2 \equiv 2$ and $2 \sum k_i^2 \equiv 2 \pmod 4$, so the left side is $\equiv 0 \pmod 4$, a contradiction. So all d_i are even and all k_i odd. Then $\sum d_i^2 \in \{0, 8\}$ and $\sum k_i^2 \in \{11, 7\}$; sums of three odd squares are $3, 11, 19, \dots$, so $d = 0$ and $\{k_i^2\} = \{9, 1, 1\}$, $k = (-3, 1, 1)$ up to order. Thus $x = y$ is a permutation of $(0, 2, 2)$, and $C_{ma_1} \in \{0, 2\}$.

Case 4. $G^w = 8(3I - J)$. In G^s we have $s_2 = s_3$ (since $|s_2 - s_3|^2 = 0$), so $n_2 = n_3$, and $s_1,\ s_1 + 2s_2$ is an orthogonal basis of S (norms $8, 24$). Also $v_m \cdot (s_1 + 2s_2) = \sum e_i = 6\sigma - 24$ with $\sigma = \sum x_i = \sum y_i$. Hence

$$3\sum d_i^2 + e_1^2 + 12(\sigma - 4)^2 = 64.$$

From $n_2 = n_3$ we get $d_2 \equiv d_3$, so $d_1 = -(d_2 + d_3)$ and n_1 are even. Thus $n_1 \in \{0, 2, 4\}$, since $e_1^2 \le 64$. Note $\sum d_i^2 \in \{0, 8, 24, \dots\}$ when all d_i are even, and $\in \{2, 6, 14, \dots\}$ otherwise.

- $n_1 = 0$: $d = 0$ and $\sigma = 4$, so $x = y = (0, 2, 2)$.
- $n_1 = 2$: $\sum d_i^2 + 4(\sigma - 4)^2 = 20$ needs $\sum d_i^2 \equiv 0 \pmod 4$, so $\sum d_i^2 \in \{0, 8\}$ and $(\sigma - 4)^2 \in \{5, 3\}$, which is impossible.
- $n_1 = 4$: $\sum d_i^2 + 4(\sigma - 4)^2 = 16$ gives $d = 0$ and $\sigma \in \{2, 6\}$, so $x = y \in \{(2, 0, 0), (2, 2, 2)\}$.

So $C_{ma_1} \in \{0, 2\}$.

Case 5. W has the orthogonal basis $w_1,\ w_2 - w_3$ (norms $16, 24$). In S, $s_2 = s_3$ (so $n_2 = n_3$), and $s_1,\ s_1 + 2s_2$ is an orthogonal basis (norms $8, 48$). With σ as in Case 4,

$$9d_1^2 + 6(d_2 - d_3)^2 + 2e_1^2 + 12(\sigma - 4)^2 = 128.$$

Again d_1 and n_1 are even, and $2e_1^2 \le 128$ forces $n_1 \in \{0, 2, 4\}$. Here $d_2 - d_3$ is even.

- $n_1 = 0$: $d = 0$ and $\sigma = 4$, so $x = y = (0, 2, 2)$.
- $n_1 = 2$: $3d_1^2 + 2(d_2 - d_3)^2 + 4(\sigma - 4)^2 = 40$ with $d_1 \in \{0, \pm 2\}$ and $(d_2 - d_3)^2 \in \{0, 4, 16\}$. No case gives a square $(\sigma - 4)^2$.
- $n_1 = 4$: $3d_1^2 + 2(d_2 - d_3)^2 + 4(\sigma - 4)^2 = 32$. The only solution is $d_1 = 0$, $d_2 - d_3 = \pm 4$, $\sigma = 4$, which gives $(x; y) \in \{(2, 2, 0; 2, 0, 2), (2, 0, 2; 2, 2, 0)\}$.

So $C_{ma_1} \in \{0, 2\}$.

Case 6. G^w has rank 1: $w_2 = w_3 = -\frac{1}{2}w_1$ and $|w_1|^2 = 16$. So $d_2 = d_3 = -d_1/2$ and $|\pi_{\mathsf{W}} v_m|^2 = \frac{9}{16}d_1^2$. G^s has rank 3, with orthogonal basis $s_1,\ 2s_2 + s_1,\ s_3 - s_1$ (norms $8, 72, 12$). Hence

$$9d_1^2 + 2(3n_1 - 8)^2 + 2(n_1 + 2n_2 - 8)^2 + 12(n_3 - n_1)^2 = 128.$$

Here d_1 is even and $|d_1| \le 2$, since $9 \cdot 16 > 128$.

- $d_1 = 0$: then $x = y$, $n = 2x$, and the equation becomes $(3x_1 - 4)^2 + (x_1 + 2x_2 - 4)^2 + 6(x_3 - x_1)^2 = 16$. So $x_1 \le 2$. The case $x_1 = 0$ gives $x = (0, 2, 0)$. The case $x_1 = 1$ needs $(2x_2 - 3)^2 = 9$ and $(x_3 - 1)^2 = 1$, giving $x \in \{(1, 0, 0), (1, 0, 2), (1, 3, 0), (1, 3, 2)\}$. The case $x_1 = 2$ needs $2(x_2 - 1)^2 + 3(x_3 - 2)^2 = 6$, which is impossible.
- $d_1 = \pm 2$: then n_1 is even, n_2, n_3 are odd, and $(3n_1 - 8)^2 + (n_1 + 2n_2 - 8)^2 + 6(n_3 - n_1)^2 = 46$ with $n_3 - n_1$ odd and $n_1 \in \{2, 4\}$. For $n_1 = 2$ it forces $(2n_2 - 6)^2 = 36$, so n_2 is even, a contradiction. For $n_1 = 4$ it forces $(2n_2 - 4)^2 = 24$, which is impossible.

So $x = y$ and $C_{ma_1} \in \{0, 1\}$.

Conclusion. Now look at row a_1 of C restricted to the six mixed columns. By (6), its entries sum to $12 - 2\sigma_{a_1}$, and their squares sum to $22 - \sum_{j \in \mathcal{A} \cup \mathcal{B}} C_{a_1 j}^2$:

case	possible $C_{a_1 m}$	$\sum_m C_{a_1 m}$	$\sum_m C_{a_1 m}^2$	contradiction
1	$\{0, 1, 3\}$	8	16	$n_1 + 3n_3 = 8$, $n_1 + 9n_3 = 16 \Rightarrow 6n_3 = 8$
2	$\{0, 1\}$	8	16	six entries in $\{0, 1\}$ sum to at most 6
3	$\{0, 2\}$	8	18	$c^2 = 2c$ forces square-sum 16
4	$\{0, 2\}$	8	18	same
5	$\{0, 2\}$	8	18	same
6	$\{0, 1\}$	6	12	$c^2 = c$ forces square-sum 6

(In case 1, n_1, n_3 count the entries equal to 1 and 3.) Every case is contradictory, so no matrix C as in Propositions 4.1 and 5.2 exists. This proves Theorem 1.1. □

Remark 8.2. The final step needs only the values C_{ma_1}. The full lists of mixed vectors (six, four, three, three, three and five of them) are recorded because they are used in the formal proof.

9. Independent verification and formalization

The scripts are in `ordre7/scripts` of the repository. All of them use exact arithmetic.

- `census6.py` enumerates all 6×6 blocks with entries in $[0,4]$ that satisfy (11), row feasibility, and positive semidefiniteness of rank ≤ 4 of the Gram block. It finds 8 classes, and exactly the six of Theorem 7.5 once $h \le 5$ is imposed.
- `mixed6b.py` recomputes the mixed vectors of Section 8 by testing all $x \in [0,4]^6$ against the full 7×7 Gram condition (positive semidefinite of rank 4), together with the contradictions of the final table. `residuals.py` reproduces the tables of Section 7.
- `fullsearch.c` is a brute-force search with no Gram reasoning, over all symmetric 12×12 matrices with entries in $[0,4]$ satisfying (2)–(4). With zero diagonal it finds no solution (about $8 \cdot 10^9$ nodes). It also finds none for each of the nine diagonal patterns with $C_{ii} \in \{0,2\}$ and $\operatorname{tr} C = 14$, the only other traces compatible with Lemma 4.2 and the local structure. This suggests that the representation-theoretic step could be avoided; it is not needed for the proof.

Lean. The Lean 4/Mathlib development is in `ordre7/lean`; its status is described in `ordre7/lean/README.md`.

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